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I'm a Postdoctoral researcher at the Technical University of Denmark working at the intersection of transportation science, climate adaptation, and urban sustainability. My research integrates spatial analytics, machine learning, and environmental modeling to understand human-environment interactions in urban contexts, with particular focus on climate risks, road safety, mobility, and wellbeing. I'm currently developing an AI-driven spatial framework to model how flooding impacts urban transportation systems and community quality of life, and to discover optimal adaptation pathways through reinforcement learning. I obtained my PhD (*summa cum laude*) from Instituto Superior Técnico, Portugal, where I developed geospatial machine learning methods to map cycling safety, both objective accident risk and subjective safety perceptions.

Research Interests: Intelligent Transportation Systems, Urban Environments & Contexts, Geospatial Analysis, Smart Mobility, Computer Vision, Machine Learning, Urban Mobility, Climate Adaptation, GeoAI

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One-Page Summary

Education

2024, PhD **Transportation Systems (MIT Portugal)**, Instituto Superior Técnico (IST), Portugal
Pass with Distinction and Honor (equivalent to *summa cum laude*)

Current Role

Mar 2024 - Present **Postdoctoral Researcher**, Technical University of Denmark (DTU), Denmark

Highlighted Project

MA'AT Development of a **climate adaptation AI framework** that combines **climate modeling**, **transport simulation**, and **well-being** modeling for policy evaluation. Main developer/contributor and responsible on integration between **different models and domains**.

Teaching

Teaching Teacher and Guest Lecturer (DTU, *Advanced Business Analytics*)
Guest Lecturer (IST, *Transport Demand Modelling*)

Accreditation *Teaching Lab* (DTU, course for Assistant/Associate Professors)
Teaching STEM Students (IST, course for Assistant Professors and research assistants)

Supervision 4 MSc Thesis (+1 ongoing), 2 BSc Special Projects (+1 ongoing), Startup Technical advisor

Scientific Output

Publications 9 journal articles (+2 under review, +3 preprints, +3 finalizing drafts), 20 peer-reviewed conference publications

Presentations 2 keynote talks, 4 invited research talks, 16 conference presentations

Statistics 161 citations, h-index: 8 (July 28, 2026, [Google Scholar](#))

Awards Seal of Excellence (MSCA), Best Proposal Award (ICLR Workshop)

Network *Current collaborations:* Technical University of Denmark (Denmark), Instituto Superior Técnico (Portugal), University of Galway (Ireland), Polytechnic University of Catalonia (Spain), Aalto University (Finland), Delft University of Technology (Netherlands), Chalmers University (Sweden), Technical University of Munich (Germany), National University of Singapore (Singapore). *Past collaborations:* National Laboratory of Civil Engineering (Portugal), VTM (Portugal), NOVA University Lisbon (Portugal). I am also part of a wide European network of transportation researchers (EIT Urban Mobility DTN). The list includes only individuals I have worked closely with and does not include less relevant forms of contacts such as project proposals.

Scientific and Societal Impact

Impact assessment and future policy evaluation: I create models that identify how context shapes outcomes and which interventions work best. My research addresses road safety, climate impacts on transportation, and quality of life. I work closely with domain experts to build valid, practical simulation frameworks.

Strong multidisciplinary focus for responsible research, innovation, and sustainability: I pursue a responsible, multidisciplinary research agenda exploring how data-driven methods (e.g., AI) can advance geography, transport, and urban sciences. Today's challenges require interdisciplinary approaches that leverage new data modalities, computational advances, and cross-domain insights. By building simulation frameworks that cross domains, I create reliable tools for testing policies and sustainable solutions.

Open-access publications, data, and code: Most of my work has been published with open-access to enhance accessibility and impact. I also openly share the data and code used across my work, usually in my Github.

1. Education

- PhD** **Transportation Systems (MIT Portugal), 2024**
Instituto Superior Técnico, Universidade de Lisboa, Portugal
Pass with Distinction and Honor (equivalent to *summa cum laude*)
- MSc & BSc** **Electrical & Computer Engineering, 2017**
Instituto Superior Técnico, Universidade de Lisboa, Portugal

2. Experience

- Mar 2024 - Present** **Technical University of Denmark, Denmark**
Postdoctoral Researcher
Developing tools that aid climate change adaptation planning using a multi-modular AI framework to model dynamics effects between climate, transport, and wellbeing.
Mentor: Francisco C. Pereira
- Oct 2018 – Feb 2024** **Instituto Superior Técnico, Portugal**
Graduate Researcher
Developed a framework using machine learning for understanding and mapping objective cycling safety, subjective cycling safety, and the relation between the two.
Supervisors: Filipe Moura, Manuel Marques, and Carlos Lima Azevedo
- May 2022 – Dec 2022** **Technical University of Denmark, Denmark**
Visiting Researcher
Analysed objective cycling safety using a machine learning-econometrics combined model. Created a novel framework for studying subjective cycling safety.
Hosts: Carlos Lima Azevedo and Felix Wilhelm Siebert
- Jul 2017 - Sep 2018** **Instituto Superior Técnico, Portugal**
Research Assistant
Created 3 data acquisition platforms: i) 3D scanning sidewalks, ii) cycling maneuvers recognition; iii) counting passengers in a subway trains.
Supervisors: Manuel Marques and João Paulo Costeira

3. Teaching

- 2026–** **(Spring)** **Technical University of Denmark, Denmark**
Teacher
42578 Advanced Business Analytics (graduate level, MSc)
Designed and presented modules on *Computer Vision* and *Reinforcement Learning*, including theoretical concepts and hands-on coding examples (notebook-style code). Evaluated final group projects.
- 2024–2025 (Spring)** **Technical University of Denmark, Denmark**
Guest Lecturer
42578 Advanced Business Analytics (graduate level, MSc)
Designed and presented a module on *Introduction to Computer Vision*, including theoretical concepts and practical examples (notebook-style code).

2019–2023 (Fall)

Instituto Superior Técnico, Portugal

Guest Lecturer

Transport Demand Modelling (graduate level, PhD)

Designed and presented a module on *Discrete Choice Models*, including theoretical concepts, algorithmic details, and practical examples (notebook-style code). Designed and graded final projects.

4. Publications

Google Scholar: <https://scholar.google.com/citations?user=kiSlzroAAAAJ&hl=en>

Scopus: <https://www.scopus.com/authid/detail.uri?authorId=58828745900>

Web of Science: <https://www.webofscience.com/wos/author/record/HNP-2151-2023>

Selected Publications In Preparation

3. Vandervoort, A., **Costa, M.**, Drews, M., Pereira, F. C., Morrissey, K. “Using transport justice to estimate the distributional effects of nuisance and emergency flooding in Denmark’s five most populous cities”
2. Vandervoort, A., **Costa, M.**, Drews, M., Pereira, F. C., Morrissey, K. “Nuisance Flooding Lowers Quality of Life: a Copenhagen Case Study”
1. Siebert, F. W., Riis, C., Hüttel, F. B., **Costa, M.** “Scaling Motorcycle Helmet Use Registration by Applying Computer Vision to Crowd-Sourced Images”

Publications Under Review and Preprints

5. **Costa, M.**, Vandervoort, A., Drews, M., Morrissey, K., Pereira, F. C. “Transport Impacts and Adaptation Strategies for Pluvial Flooding in Copenhagen: Economic Costs to Driving, Cycling and Walking”. Under review in *Transportation Research Part D: Transport and Environment*. (2026)
4. **Costa, M.**, Azevedo, C. L., Siebert, F. W., Marques, M., Moura, F. “Feeling Safe vs. Being Safe: Comparing Objective and Subjective Cycling Safety in Urban Contexts Using Machine Learning”. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6872979. (2026)
3. **Costa, M.**, Vandervoort, A., Schmidt, C., Petersen, M. W., Drews, M., Morrissey, K., Pereira, F. C. “Learning long term climate-resilient transport adaptation pathways under direct and indirect flood impacts using reinforcement learning”. <https://arxiv.org/abs/2601.18586>. (2026)
2. **Costa, M.**, Siebert, F. W., Azevedo, C. L., Marques, M., Moura, F. “Understanding perception of cycling safety from street-view images: uncovering non-linear relations to urban factors”. <https://dx.doi.org/10.2139/ssrn.4886041>. (2024)
1. Christ, A. K., **Costa, M.**, Marques, M., Roque, C., Moura, F. “A Systematic Literature Review on Subjective and Objective Cycling Safety Measures”. <https://dx.doi.org/10.2139/ssrn.4886041>. Under review in *Journal of Safety Research* (2024)

Journal Articles

- 2026 Wang, X., Kovaceva, J., **Costa, M.**, Pereira, F. C., Thompson, R. “Domain-Adapted Pre-trained Language Models for Implicit Information Extraction in Crash Narratives”. *Transportation Research Part C: Emerging Technologies* (in print). <https://arxiv.org/abs/2510.09434>.

- 2026 Ito, K., Terra, R., **Costa, M.**, Siebert, F. W., Garrido-Valenzuela, F., Azevedo, C. L., Biljecki, F., van Cranenburgh, S. “Perceived Traffic Safety’s Impact on Cycling Route Choice with Street View Images.” *Transportation*. <https://doi.org/10.1007/s11116-026-10793-z>.
- 2025 **Costa, M.**, Marques, M., Siebert, F. W., Azevedo, C. L., Moura, F. “Bridging the Gap: A Scoping Review Exploring the Relationship Between Objective and Subjective Cycling Safety.” *Transport Reviews*, 45(6), 993–1015. <https://doi.org/10.1080/01441647.2025.2537200>.
- 2025 **Costa, M.**, Valença, G., Adorean, C., Moura, F. “The Relation between Circuity and Mobility Cultures: A Study of 41 European Cities.” *Cities*, 166, 106087. <https://doi.org/10.1016/j.cities.2025.106087>.
- 2025 **Costa, M.**, Marques, M., Azevedo, C. L., Siebert, F. W., Moura, F. “Which cycling environment appears safer? Learning cycling safety perceptions from pairwise image comparisons.” *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 2. <https://doi.org/10.1109/TITS.2024.3507639>.
- 2024 **Costa, M.**, Azevedo, C. L., Siebert, F. W., Marques, M., Moura, F. “Unraveling the Relation between Cycling Accidents and Built Environment Typologies: Capturing Spatial Heterogeneity through a Latent Class Discrete Outcome Model.” *Accident Analysis & Prevention*, 200, 107533. <https://doi.org/10.1016/j.aap.2024.107533>.
- 2022 **Costa, M.**, Marques, M., Roque, C., Moura, F. “CYCLANDS: Cycling geo-located accidents, their details and severities”. *Scientific Data*, 9, 237. <https://doi.org/10.1038/s41597-022-01333-2>.
- 2022 **Costa, M.**, Félix, R., Marques, M., Moura, F. “Impact of COVID-19 lockdown on the behavior change of cyclists in Lisbon, using multinomial logit regression analysis.” *Transportation Research Interdisciplinary Perspectives*, 14, 100609. <https://doi.org/10.1016/j.trip.2022.100609>.
- 2021 **Costa, M.**, Marques, M., Moura, F. “A Circuity Temporal Analysis of Urban Street Networks Using Open Data: A Lisbon Case Study.” *ISPRS International Journal of Geo-Information*, 10(7), 453.

Peer-Reviewed Conference Articles*

- 2026 **Costa, M.**, Azevedo, C. L., Siebert, F. W., Marques, M., Moura, F. “Comparing Urban Elements Effects on Objective and Subjective Cycling Safety”. Accepted for publication in *14th Symposium of the European Association for Research in Transportation (hEART 2026)*. Paris, France.
- 2026 Madaleno, F., Lassen, O. B., Agriesti, S., **Costa, M.**, Azevedo, C. L., Rodrigues, F., Pereira, F. C. “Interventionally Consistent Surrogates for Urban Flood Resilience Planning”. Accepted for publication in *14th Symposium of the European Association for Research in Transportation (hEART 2026)*. Paris, France.
- 2026 Vandervoort, A., **Costa, M.**, Drews, M., Pereira, F. C., Morrissey, K. “How can nuisance flooding affect Copenhagen residents’ Quality of Life?”. Accepted for publication in *14th Symposium of the European Association for Research in Transportation (hEART 2026)*. Paris, France.
- 2026 Perdigão, L. M., **Costa, M.**, Santiago, C., Marques, M. “Learning to Look Like Humans: Gaze-Aligned Cycling Safety Prediction”. Accepted for publication in *IEEE Intelligent Transportation Systems Conference 2026*. Naples, Italy.
- 2025 **Costa, M.**, Vandervoort, A., Drews, M., Morrissey, K., Pereira, F. C. “Climate Adaptation with Reinforcement Learning: Economic vs. Quality of Life Adaptation Pathways.” *AI for Climate and Conservation Workshop at EurIPS 2025*. Copenhagen, Denmark.
- 2025 **Costa, M.**, Vandervoort, A., Drews, M., Morrissey, K., Pereira, F. C. “Incorporating Quality of Life in Climate Adaptation Planning via Reinforcement Learning.” *AI in Science (AIS) 2025*. Copenhagen, Denmark
- 2025 Wang, X., **Costa, M.**, Kovaceva, J., Wang, S., Pereira, F. C. “Plugging Schema Graph into Multi-Table QA: A Human-Guided Framework for Reducing LLM Reliance.” *Findings of the Association for Computational Linguistics: EMNLP 2025*. pp. 5829–5842. Association for Computational Linguistics. Suzhou, China.

* Presenter *italicized* if not first author; (*) denotes shared first-authorship.

- 2025 Vandervoort, A. *, **Costa, M. ***, Petersen, M. W., Drews, M., Haustein, S., Morrissey, K., Pereira, F. C. “Using Reinforcement Learning to Integrate Subjective Wellbeing into Climate Adaptation Decision Making.” *Tackling Climate Change with Machine Learning workshop at International Conference on Learning Representations (ICLR) 2025*. Singapore. **Best Proposal Award**
- 2024 **Costa, M.**, Petersen, M. W., Vandervoort, A., Drews, M., Morrissey, K., Pereira, F. C. “Climate Adaptation with Reinforcement Learning: Experiments with Flooding and Transportation in Copenhagen.” *Tackling Climate Change with Machine Learning workshop at Conference on Neural Information Processing Systems (NeurIPS) 2024*. Vancouver, Canada.
- 2024 **Costa, M.**, Marques, M., Moura, F., Azevedo, C. L., Siebert, F. W. “Comparing the Effects of Urban Elements on Objective and Subjective Cycling Safety: A Berlin Case Study.” *International Cycling Safety Conference 2024*. Imabari, Japan.
- 2024 Siebert, F. W., Riis, C., **Costa, M.**, Hüttel, F. B. “Urban Motorcycle Helmet Use Surveillance in Southeast Asia — A Cost-Efficient Computer Vision Approach.” *15th World Conference on Injury Prevention and Safety Promotion (Safety 2024)*. New Delhi, India.
- 2024 **Costa, M.**, Azevedo, C. L., Siebert, F. W., Marques, M., Moura, F. “Capturing Spatial Heterogeneity in Cycling Accidents using a Latent Class Discrete Outcome Model.” *hEART 2024: 12th Symposium of the European Association for Research in Transportation*. Helsinki, Finland.
- 2023 **Costa, M.**, Marques, M., Siebert, F. W., Azevedo, C. L., Moura, F. “Understanding Perception of Cycling Safety through Pairwise Image Comparisons.” *International Cycling Safety Conference 2023*. The Hague, The Netherlands.
- 2023 **Costa, M.**, Marques, M., Siebert, F. W., Azevedo, C. L., Moura, F. “Scoring Cycling Environments Perceived Safety using Pairwise Image Comparisons.” *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE. pp. 4046-4051. Bilbao, Spain.
- 2022 **Costa, M.**, Roque, C., Marques, M., Moura, F. “Cycling Safety Data Augmentation in the Urban Environment: A Barcelona Case Study.” *International Cycling Safety Conference 2022*. Dresden, Germany.
- 2022 Christ, A. K., **Costa, M.**, Marques, M., Roque, C., Moura, F. “Perceiving objective cycling safety: a systematic literature review.” *Transportation Research Arena*. Lisboa, Portugal.
- 2021 **Costa, M.**, Félix, R., Marques, M., Moura, F. “Impact of COVID-19 lockdown on the Behavior Change of Cyclists in Lisbon, using multinomial logit regression analysis.” *Transportation Research Board Annual Meeting 2021*. (Online) Washington DC, United States of America.
- 2019 **Costa, M.**, Cambra, P., Marques, M., Moura, F. “WalkBot: A Portable System to Scan Sidewalks.” *2019 IEEE International Smart Cities Conference (ISC2)*. (pp. 167-172). IEEE. Casablanca, Morocco.
- 2019 **Costa, M.**, Cambra, P., Marques, M., Moura, F. “WalkBot – Desenvolvimento de um Equipamento de Avaliação da Qualidade da Infra-estrutura Pedonal com Recurso a Processamento de Imagem Tridimensional.” *Congresso Rodoviário Português*. Lisbon, Portugal.
- 2017 **Costa, M.**, Ferreira, B. Q., Marques, M. “A context aware and video-based risk descriptor for cyclists.” *2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC)*. (pp. 1-6). IEEE. Yokohama, Japan.

Thesis

- 2024 “Objective and Subjective Safety Mapping for Urban Cyclists.” Ph.D. Thesis, Transportation Systems, Instituto Superior Técnico, University of Lisbon
- 2017 “Video-Based Risk Assessment for Cyclists.” M.Sc. Thesis, Electrical and Computer Engineering, Instituto Superior Técnico, University of Lisbon

Teaching Materials

- 2021 Moura, F., Valença, G., **Costa, M.**, Roque, C., Félix, R. “U-Shift/Transport-Demand-Modelling: Supporting materials to Transportation Demand Modelling classes at Instituto Superior Técnico - University of Lisbon. (Version 2021.0)”. *GitHub*. <http://doi.org/10.5281/zenodo.4599525>

5. Conference Activity & Invited Talks

Keynote and Plenary Addresses

- 2025 “Objective and Subjective Safety Mapping for Urban Cyclists.” *10th International Conference on Intelligent Transportation Engineering (ICITE) 2025*. Beijing, China, Oct 24–26.
- 2025 “Switching Gears: Pedalling to Climate Resilient Transport.” *EIT Doctoral Training Network Annual Forum 2025*. Lisbon, Portugal, Sep 10–12.

Invited Research Talks

- 2026 “Artificial Intelligence for Climate Adaptation: Reinforcement Learning for Climate Change-Resilient Transport.” *IVe-Seminar, Institute for Transport Studies, BOKU University*. Vienna, Austria, Apr 20.
- 2026 “Feeling Safe vs. Being Safe: Comparing Objective and Subjective Cycling Safety in Urban Contexts.” *Active Mobility Workshop, Masaryk University*. Brno, Czech Republic, Apr 17.
- 2025 “Feeling Safe vs. Being Safe: Comparing Objective and Subjective Cycling Safety in Urban Contexts.” *GRETTIA, Université Gustave Eiffel*. Paris, France, Oct 16.
- 2024 “Which environment looks safer? Using machine learning for analyzing perception of cycling safety in Berlin.” *Big data and Transport, University of Cape Town*. Cape Town, South Africa (online), Oct 24.

Sessions Chaired/Moderated

- 2025 “Collaborative Control and Scheduling of Multiple Transportation Modes in Intelligent Transportation Networks.” *10th International Conference on Intelligent Transportation Engineering (ICITE) 2025*. Beijing, China, Oct 24–26.
- 2024 “Pitch your Research.” *EIT Doctoral Training Network Annual Forum 2024*. Ghent University. Ghent, Belgium, Sep 16–18.

Conference Presentations (not peer-reviewed)[†]

- 2026 **Costa, M.**, Marques, M., Azevedo, C. L., Siebert, F. W., Moura, F. “Which cycling environment appears safer? Learning cycling safety perceptions from pairwise image comparisons.” Accepted for presentation in 2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC), Journal Presentations track. Naples, Italy.
- 2026 **Costa, M.**, Vandervoort, A., Drews, M., Morrissey, K., Pereira, F. C. “MA’AT: Maximising wellbeing with AI under deep climate turmoil.” *EIT Urban Mobility Community Days 2026*. Copenhagen, Denmark (Poster)
- 2025 **Costa, M.**, Vandervoort, A., Drews, M., Morrissey, K., Pereira, F. C. “Climate Adaptation with Reinforcement Learning: Economic vs. Quality of Life Adaptation Pathways.” *Danish Meteorological Institute’s 2025 Climate Symposium*. Copenhagen, Denmark (Poster)

[†] Presenter *italicized* if not first author.

- 2024 Siebert, F. W., Riis, C., **Costa, M.**, Hüttel, F. B. “Urban Motorcycle Helmet Use Surveillance in Southeast Asia - A Cost-Efficient Computer Vision Approach.” *15th World Conference on Injury Prevention and Safety Promotion (Safety 2024)*. New Delhi, India (Oral)
- 2023 **Costa, M.**, Siebert, F. W., Marques, M., Azevedo, C. L., Moura, F. “Understanding cyclists’ perception of safety using eye tracking.” *Grupo de Estudos de Transportes 2023*. Oeiras, Portugal (Oral)
- 2022 **Costa, M.**, Marques, M., Siebert, F. W., Azevedo, C. L., Moura, F. “Subjective Risk Mapping for Urban Cyclists.” *EIT Doctoral Training Network Annual Forum 2022*. Barcelona, Spain (Oral)
- 2022 **Costa, M.**, Marques, M., Siebert, F. W., Azevedo, C. L., Moura, F. “Subjective Risk Mapping for Urban Cyclists.” *Grupo de Estudos de Transportes 2022*. Porto, Portugal (Oral)
- 2022 Christ, A. K., **Costa, M.**, Marques, M., Roque, C., Moura, F. “Percebendo a segurança objetiva dos ciclistas urbanos: uma revisão sistemática da literatura.” *10º Congresso Rodoferroviário Português*. Lisboa, Portugal (Oral)
- 2022 **Costa, M.**, Marques, M., Moura, F. “Como é que as redes rodoviárias, pedonais e cicláveis mudam ao longo do tempo? Uma análise da rectilinearidade em Lisboa, Portugal.” *10º Congresso Rodoferroviário Português*. Lisboa, Portugal (Oral)
- 2021 **Costa, M.**, Marques, M., Moura, F. “Objective and Subjective Risk Mapping for Urban Cyclists.” *EIT Doctoral Training Network Annual Forum 2021*. Munich, Germany (Oral)
- 2020 **Costa, M.**, Marques, M., Moura, F. “Objective and Subjective Risk Mapping for Urban Cyclists.” *EIT Doctoral Training Network Annual Forum 2020*. (Online)
- 2019 **Costa, M.**, Cambra, P., Marques, M., Moura, F. “Bike Rider - Sensorial Mapping of Cyclable Paths.” *Velo-City 2019 ‘Cycling for the Ages’*. Dublin, Ireland (Oral)
- 2018 **Costa, M.**, Cambra, P., Marques, M., Moura, F. “WalkBot: A Portable System to Scan Sidewalks.” *LARSyS 2018 Annual Meeting*. Lisbon, Portugal (Oral)
- 2018 **Costa, M.**, Marques, M., Moura, F. “Smartbike: Cyclists’ Perception of Risk.” *LARSyS 2018 Annual Meeting*. Lisbon, Portugal (Oral)
- 2017 **Costa, M.**, Marques, M., Moura, F. “A Context Aware and Video-Based Risk Descriptor for Cyclists.” *Ciência 2017 – Meeting with Science and Technology*. Lisbon, Portugal (Poster)
- 2017 **Costa, M.**, Marques, M., Moura, F. “Mobility in Cities: Assessing Experiences Using a Smartphone.” *Ciência 2016 – Meeting with Science and Technology*. Lisbon, Portugal (Poster)

6. Research Grants and Funding

Awarded

Awarded

- 2026 **Co-PI & Knowledge Hub Ambassador.** *GLOWB: Green Liveability & mobility for WellBeing*. Driving Urban Transitions (DUT) Partnership. Total: 1.164.769,75 € (own share: 299.998,00 €). Awarded.

Submitted / Under Review

- 2026 **Co-PI.** *PROTEUS: Predictive Robust Operational Twins for Environmental Uncertainty Simulation*. Google.org Impact Challenge: AI for Science. (3.000.000 \$). Under Review.
- 2026 **Named Researcher.** *CARA: Climate Adaptation and Resilient cities through AI*. Villum Fonden Synergy Shortlisted after Stage 1; invited to Stage 2 interview.

Selected Past Applications

- 2026 **Lead Applicant.** *FUTURBAN: Learning Multimodal Spatial and Temporal Representations of Urban Dynamics for Future Planning*. Postdoctoral Fellowship, Danish Data Science Academy (1.365.000 DKK). Ranked 8th out of 85. Not funded.
- 2025 **Lead Applicant.** *FUTURBAN – Learning Multimodal Spatial and Temporal Representations of Urban Dynamics for Future Planning*. Marie Skłodowska-Curie Postdoctoral Fellowship, European Commission (226.420,56€). Overall score: 94.6/100% — Not funded.
Awarded the Seal of Excellence.
- 2025 **Co-author** (Partner PI: Francisco C. Pereira, DTU). *ATHENEA – An AI-Based Digital Twin For Shaping, Planning, and Managing Climate-Neutral Urban Futures*, Consortium led by Universidad Politécnica de Madrid. EU Horizon HORIZON-MISS-2025-04-CIT-02 (European Commission) (903.372,45€). Author of reinforcement learning component for long-term climate planning work package. Overall evaluation score: 13/15 — Not funded.
- 2020 **Primary Author / Contributor** (PI: Dr. Rosa Félix). “*SAFECYCLING – Automatic, Continuous and Ubiquitous Mapping of Cyclists’ Subjective Safety*”, Fundação para a Ciência e Tecnologia (Portuguese national research funding agency) (249.792,18€). Led proposal development, including research design, methodology, and budget preparation. Overall evaluation score: 7.8/9.0 (Scientific Merit: 8.25/9.0) — Not funded.
- 2018 **Contributor** (PI: Prof. Filipe Moura). “*BikeRider: Sensory navigation system adapted to different cyclist profiles*”, 2nd Edition of TecInnov Thales — IST (3.820€). Contributed to methodology development. Funded Jan 2018.
- 2017 **Contributor** (PI: Prof. Filipe Moura). “*WalkBot: Automated collection of urban pedestrian accessibility data*”, 1st Edition of TecInnov Thales — IST (4.970€). Contributed to methodology development. Funded Aug 2017.

7. Honors & Awards

- 2026 **Seal of Excellence** *Marie Skłodowska-Curie Actions, Postdoctoral Fellowship*.
- 2025 **Best Proposal Award** *Tackling Climate Change with Machine Learning workshop, International Conference on Learning Representations (ICLR) 2025*.
- 2025 **Travel Grant** *Otto Mønstedts Fond*. 7.500 DKK.
- 2024 **Travel Grant** *Otto Mønstedts Fond*. 7.500 DKK.
- 2024 **PhD awarded with Distinction and Honor** *Instituto Superior Técnico* (highest doctoral distinction, equivalent to *summa cum laude*).
- 2018 **MIT Portugal Doctoral Fellowship (4-year PhD grant)** *Fundação para a Ciência e Tecnologia* (national science funding institution in Portugal).

8. Supervision & Mentoring

MSc Supervisor

1. Rita Saraiva (DTU), Ongoing.
2. Beatriz Braga de Carvalho (DTU), 2026. *Analysing Reinforcement Learning Policy Behavior for Climate Adaptation: A Study within the MA’AT Framework*
3. João Machado Cerejo de Miranda (DTU), 2026. *Sample Efficient Climate Adaptation Policy Optimization: A Bayesian Optimization Benchmark for the MA’AT Framework*
4. Xixi Wang (DTU, with Chalmers University of Technology), 2025. *AI-Driven Analysis of Crash Narratives: Evaluating Open-Source LLMs for Structured Information Extraction*
→ Work presented at the Empirical Methods in Natural Language Processing (EMNLP 2025) (A* AI Conference)
→ Work submitted to Transportation Research Part C: Emerging Technologies.

5. **Morten William Petersen (DTU), 2024.** *Deep Reinforcement Learning adds value to an Integrated Assessment Model for guiding policymakers*
→ Work presented at the Tackling Climate Change with Machine Learning workshop at the Conference on Neural Information Processing Systems (NeurIPS) 2024 (A* AI Conference)

BSc & Special Project Supervisor

1. **Theresa Bartels, Max Bürkner, Jonathan Aastrup Hoch, Gerard Ramos Pomar (DTU), Ongoing.** *Climate action with Reinforcement Learning*
2. **Rita Saraiva (DTU), 2026.** *Reinforcement Learning til vurdering af oversvømmelsespåvirkning i Storkøbenhavn*
3. **Emilie Grønberg Kristensen (DTU), 2025.** *Climate adaptation and vulnerability of urban transport systems to flooding.*

Industry & Start-ups

2021–2022 **Dilemma Insights - Technical Advisor** *Providing technical consultancy on how to use business game simulations within a classroom environment.*

9. Media & Outreach

Media

2026 **AI Climate Nordics Newsletter.** *Featured profile (April).*

Outreach

2019 **TreeTree2 Monitor.** *Guiding high-school students through scientific article reading and scientific process.*

10. Service

Academic Journal Peer Review

Big Earth Data
Findings
Journal of Transport & Health
Sustainable Cities and Society
Traffic Injury Prevention
Transactions on Intelligent Transportation Systems
Transactions on Intelligent Vehicles
Transportation Research Part A: Policy and Practice
Transportation Research Part F: Traffic Psychology and Behaviour
Transportation Research Record

Academic Conference Peer Review

IEEE Intelligent Transportation Systems Conference (IEEE ITSC)
Symposium of the European Association for Research in Transportation (hEART)
Transportation Research Board Annual Meeting (TRB AM)

Service to the Field

2024– **ELLIS PhD Program**, *Admissions evaluator*

11. Memberships

Climate AI Nordics, *Affiliate*

IEEE Intelligent Transportation Systems Society, *Member*

12. Extracurricular Courses & Schools

2025 **Teaching Lab.** Technical University of Denmark.

2020–2023 **EIT Urban Mobility Doctoral Training Network.** EIT & Technical University of Munich.

2019 **Teaching STEM Students.** Instituto Superior Técnico.

2015 **2015 CERN Spring Campus.** CERN & Instituto Superior Técnico.

13. References

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